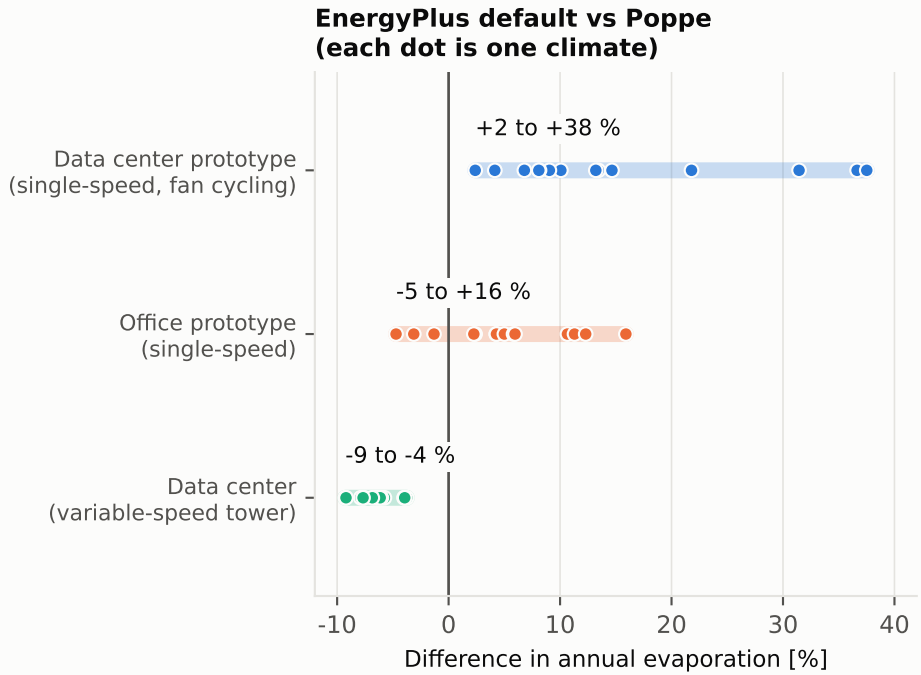
Method

8 evaporation models, from rules of thumb to the Poppe model, evaluated at the same simulated operating point every timestep

Open-source EnergyPlus plugin; fan cycling resolved into fan-on and free-convection states

3 buildings x 14 climates



Finding

Error follows the exhaust-air state:
- hot, dry air leaves unsaturated: saturated-exit models overpredict
- cold air leaves supersaturated: they underpredict

The tower's air-to-water ratio (its part-load control) decides which regime dominates